



Olaparib as first line in *BRCA*-mutated advanced ovarian carcinoma: Is it cost-effective in Spain?

Carlota Moya-Alarcón ^{a,*}, Almudena González-Domínguez ^b, Yoana Ivanova-Markova ^b, Vicente Gimeno-Ballester ^c, Maria-Pilar Barretina-Ginesta ^d, José Alejandro Pérez-Fidalgo ^e, Andrés Redondo ^f

^a AstraZeneca Farmacéutica Spain S.A., Parque Norte. C/ Serrano Galvache, 56, 28033 Madrid, Spain

^b Weber, Calle Moreto, 17, 28014 Madrid, Spain

^c Hospital Clínico Universitario Lozano Blesa, Calle San Juan Bosco, 15, 50009 Zaragoza, Spain

^d Catalan Institute of Oncology, Dr. J. Trueta University Hospital, IdibGi, Av França s/n 17007, Girona, Spain

^e University Hospital Clinic of Valencia, Av. de Blasco Ibáñez, 17, 46010, Valencia, Spain

^f Department of Medical Oncology, La Paz University Hospital-IdiPAZ, Paseo de la Castellana, 261, 28046 Madrid, Spain

HIGHLIGHTS

- Our results suggest that olaparib could delay disease progression and improve OS by 2 years vs. no maintenance treatment.
- Treatment with olaparib resulted in a gain of 2.00 QALYs vs. no maintenance treatment, at an ICER of €14,653.2/QALY.
- First-line maintenance with olaparib is cost-effective in advanced HGSOC patients with *BRCA* mutations in Spain.

ARTICLE INFO

Article history:

Received 17 June 2021

Received in revised form 3 November 2021

Accepted 13 November 2021

Available online 26 November 2021

Keywords:

Ovarian carcinoma

BRCA mutations

Olaparib

SOLO1

Cost-effectiveness analysis

Cost-utility analysis

ABSTRACT

Objective. To estimate the cost-effectiveness of olaparib after being funded by the Spanish National Health Service (SNHS) as first-line monotherapy maintenance treatment in patients with advanced high-grade serous ovarian carcinoma (HGSOC) and *BRCA* mutations in Spain.

Methods. A semi-Markov model with one-month cycles was adapted to the Spanish healthcare setting, using the perspective of the SNHS, and a time horizon of 50 years. Two scenarios were compared: receiving olaparib vs. no maintenance treatment. The model comprised four health states and included the clinical results of the SOLO1 study, along with the direct healthcare costs associated with the use of first-line and subsequent treatment resources (2020 €). A discount rate of 3% was applied for future cost and quality-of-life outcomes. A probabilistic sensitivity analysis (PSA) was also carried out and a cost-effectiveness threshold of €25,000 per quality adjusted life year (QALY) was considered.

Results. The introduction of olaparib as a first-line maintenance treatment for advanced HGSOC patients with *BRCA* mutations implied a cost of €131,614.98 compared to €102,369.54 without olaparib (difference: €29,245.44), with an improvement of 2.00 QALYs (5.56 and 3.57, respectively). Therefore, olaparib is cost-effective for advanced HGSOC patients with *BRCA* mutations, with an incremental cost-effectiveness ratio of €14,653.2/QALY. The results from the PSA showed that 92.1% of the simulations fell below the €25,000/QALY threshold. The model showed that olaparib could improve the overall survival by 2 years, vs. no maintenance treatment.

Conclusions. Olaparib as first-line maintenance treatment is cost-effective in advanced HGSOC patients with *BRCA* mutations in Spain.

© 2021 The Authors. Published by Elsevier Inc. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

Abbreviations: OC, Ovarian Cancer; HGOC, High-grade ovarian cancer; TCGA, The Cancer Genome Atlas; PARPi, Poly (ADP-ribose) polymerase inhibitors; HRR, homologous recombination repair; EMA, European Medicines Agency; NCCN, National Comprehensive Cancer Network; SEOM, Spanish Medical Oncology Society; SNHS, Spanish National Health Service; QALYs, quality-adjusted life years; ICER, incremental cost-effectiveness ratio; PFSt, progression-free state; PS1, first-progression state; PS2, second-progression state; PFS, progression-free survival; OS, overall survival; HR, hazard ratio; CI, confidence interval; PLD, pegylated liposomal doxorubicin; SD, standard deviation; AIC, Akaike information criterion.

* Corresponding author at: AstraZeneca Farmacéutica Spain, S.A., Parque Norte. C/ Serrano Galvache, 56, 28033 Madrid, Spain.

E-mail addresses: spain.healthconomics@astrazeneca.com (C. Moya-Alarcón), almudena.gonzalez@weber.org.es (A. González-Domínguez), yoana.ivanova@weber.org.es (Y. Ivanova-Markova), vgimenob@salud.aragon.es (V. Gimeno-Ballester), mpbarretina@iconcologia.net (M.-P. Barretina-Ginesta), andres.redondos@uam.es (A. Redondo).

1. Introduction

Ovarian carcinoma (OC) has one of the highest mortalities of gynecologic cancer in women. In Spain, it is expected that 3645 incident patients will be diagnosed in 2020, showing a 2.7%-increase in comparison to those diagnosed last year. However, the 5-year prevalence is relatively low (10,236 patients), due to the high mortality associated with the disease (2106 cases) in 2020 [1]. Most of OC cases have been thought to arise from epithelial cells (90%) and more than 70% of them are diagnosed with high-grade (HGOC), which typically presents at advanced stages (III and IV). This subtype grows aggressively, and generally responds to initial conventional chemotherapy [2–4]. In The Cancer Genome Atlas (TCGA) study, 20% of HGOC patients have *BRCA1* or *BRCA2* mutations: germline mutations have been recorded in 9% and 8% of the cases with *BRCA1* and *BRCA2* mutations, respectively; while somatic mutations have appeared in approximately 6% of the cases (3% *BRCA1* and other 3% *BRCA2*) [5]. *BRCA1/2* testing should be offered to all women diagnosed with high-grade epithelial non-mucinous ovarian cancer, regardless of family history [4]. Somatic testing has potential diagnostic, prognostic and predictive therapeutic implications, whereas germline testing also has preventive implications for their family members [6,7]. Although germline *BRCA1/2* mutations are the most frequent, somatic *BRCA1/2* are also predictive of response to targeted agents, and somatic testing would detect additional patients that can also benefit from the use of these therapies.

From the last 20 years, the first-line treatment of patients with advanced HGOC is contingent on the combination of cytoreductive surgery followed by first-line platinum-based chemotherapy [8,9], and it has been associated with a low rate of long-term survival. Although most *BRCA*-mutated patients have no evidence of disease after platinum-based chemotherapy, close to 80% will relapse within the first three years [10]. Relapsed patients tend to receive several treatment lines administered sequentially after each tumor progression, experiencing increasingly shorter progression-free periods on each subsequent line and developing cumulative toxicities (e.g. hypersensitivity, neurotoxicity and myelotoxicity) [11,12].

Poly (ADP-ribose) polymerase inhibitors (PARPi) represent an important new class of oral personalized agents targeting cancers with defective DNA-damage repair. The administration of this inhibitors causes accumulation of DNA damage and death of tumor cells through the homologous recombination repair (HRR) system. *BRCA* mutations are the best example of the HRR system deficiency, being a marker with a high predictive response [9]. Olaparib was the first PARPi authorized by the European Medicines Agency (EMA) for the maintenance treatment of patients with platinum-sensitive relapsed HGOC with *BRCA* mutations [13].

SOLO1 double-blind, placebo-controlled, phase III trial analyzed the efficacy and safety of olaparib as maintenance treatment for two years, in patients with newly diagnosed advanced *BRCA*-mutated high grade serous (96.5% of patients) or endometrioid (5.4% of patients) ovarian cancer (HGSOC) with clinical response after platinum-based chemotherapy. The study showed that treatment with olaparib tablets (300 mg twice daily) provided a substantial benefit, with a statistically meaningful 70%-reduction in the risk of disease progression or death in comparison to placebo with consistent and known adverse events profile [13]. As a result, olaparib was authorized by EMA in June 2019 and was recommended as first-line maintenance treatment in the National Comprehensive Cancer Network (NCCN) guidelines [14], and in the recent Spanish Medical Oncology Society (SEOM) guidelines [15]. In addition, its cost is predictable, because it has a flat-fixed dose and a maximum treatment duration of 24 months [16]. Primary results were supported by the 5-year follow-up results where the benefit derived from 2 years of maintenance olaparib was sustained beyond the end of treatment and after 5 years almost half of patients were progression-free versus 20% with placebo (HR 0.33; 95% CI 0.25–0.43) [17]. The aim of this study was to estimate the cost-effectiveness of

olaparib from the perspective of the Spanish National Health Service (SNHS) as first-line monotherapy maintenance treatment in patients with advanced HGSOC and *BRCA* mutations who are in response to first-line platinum-based chemotherapy after being commercialized.

2. Materials and methods

A partitioned survival model with one-month cycles was originally developed for the United Kingdom [18] using Microsoft Excel, and was subsequently adapted to the SNHS setting. Informed by the SOLO1 trial, the model considered two scenarios: receiving olaparib as first-line maintenance treatment in advanced HGSOC *BRCA*-mutated patients vs. no maintenance treatment. The perspective was that of the SNHS and the time horizon was 50 years (covering lifetime of these patients). Future costs and quality-of-life outcomes were discounted at an annual rate of 3%, as reported by Spanish health technology assessment recommendations [19]. Model results were costs and quality-adjusted life years (QALYs), and the incremental cost-effectiveness ratio (ICER) was estimated. According to the Spanish guidelines, a cost-effectiveness threshold between €25,000 - €60,000/QALY was considered [20,21].

2.1. Data sources

Epidemiology data, utilization of healthcare resources and utilities were obtained from a literature review, including national and international references (the latter were used whenever national data were not available). Databases consulted were Medline/Pubmed, Embase, Medes, and other official databases. All extracted data were contrasted and validated by a multidisciplinary expert group composed of clinical experts and pharmacoeconomic specialists. To validate the data, each expert first answered an online survey and then attended an in-person group meeting to reach a final consensus.

2.2. Model

The model included four health states (Fig. S1): progression-free state (PFSt), first-progression state (PS1), second-progression state (PS2) and death. The simulation was initiated with patients in PFSt who received a first-line platinum-based chemotherapy, and may suffer disease relapse, progress to PS1 or die. PS1 patients may have another relapse (PS2) or die. As the progression of the disease is an irreversible process, patients in PS2 cannot return to PS1. Death is the absorbing state in the model.

2.3. Efficacy

Patients progression in PS1 and PS2 was informed by the first and second progression-free survivals (PFS) from SOLO1 trial [13]. The long-term survival of patients was estimated using Weibull distributions, fitted to the Kaplan-Meier curves from the clinical trial for first PFS, second PFS and overall survival (OS). Fig. 1 shows the predicted first and second PFS and OS curves.

Mortality rates due to HGSOC and all causes were considered. The mortality risk in stable patients and those who had a remission period longer than 10 years [8], was that of the Spanish general population [22], adjusted for the potential excess mortality risk of having a *BRCA1/2* mutation. The excess mortality risk was modeled using a hazard ratio (HR) for mortality of 1.26 (95% confidence interval [CI]: 0.00, 3.42), based on the excess mortality of female *BRCA1/2* mutation carriers, aged 51–60 years and who have an absence of melanoma and breast, ovary or/and pancreas cancer [23].

2.4. Resource use

Healthcare resource use included first-line maintenance, second-line $\pm$ maintenance and subsequent lines of treatment, adverse events, oncology visits, follow-up tests and palliative care.

The model included those drugs used post-progression after first-line maintenance therapy in the SOLO1 trial (carboplatin, gemcitabine, paclitaxel, bevacizumab, trabectedin, pegylated liposomal doxorubicin [PLD], and olaparib [no maintenance treatment scenario], along with the combinations of these drugs usually used [13] (Table S1, S2

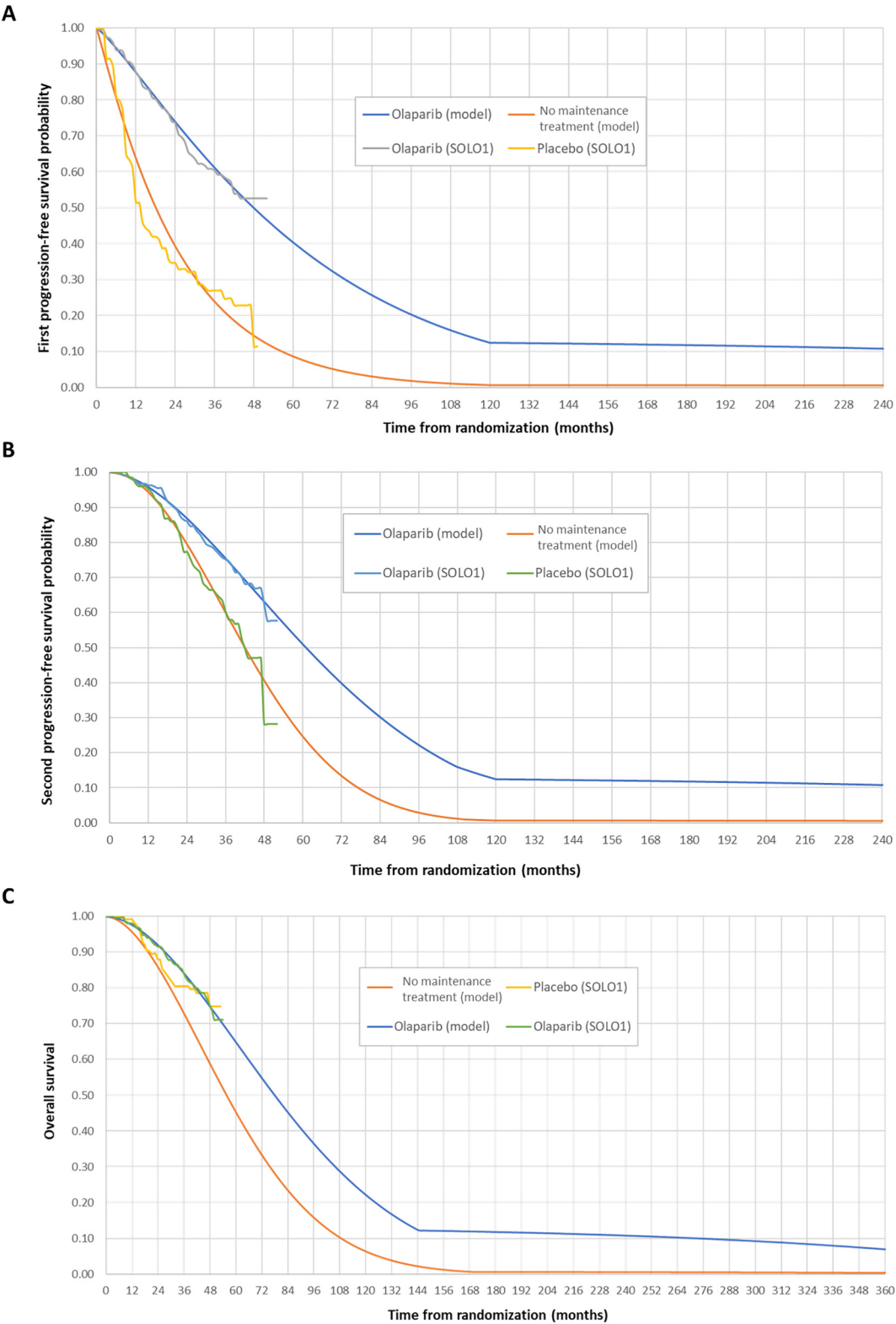


Fig. 1. Comparison of first-progression (A), second-progression (B) free survival and overall survival (C) curves from the clinical trial (SOLO1) vs. the simulation results (model).

and S3). It was considered that 56.8% of the relapsed patients received platinum-based chemotherapy for 6 cycles. The market shares of platinum and non-platinum-based chemotherapy are shown in Table S1 and S2. In the model, relapsed patients did not receive any PARPi in the olaparib scenario, because rechallenge is still being investigated [24]. However, as in the SOLO1, 39.2% of the relapsed patients received PARPi in the no maintenance treatment scenario (mainly olaparib) as a second-line maintenance treatment after platinum-based chemotherapy until the progression of the disease [25].

The model also considered the most frequent severe adverse events (grades 3 and 4) associated with olaparib and placebo in SOLO1: anemia, fatigue or asthenia, neutropenia and diarrhea (Table 1) [13].

In addition, our study included all oncology visits and follow-up tests required for patients in both scenarios (stable patients and those showing progression), based on the time interval since the administration of the first-line maintenance treatment (Table 2) [15]. Finally, it was assumed that palliative care is given to the patients in their last 48 days (terminal phase) [26].

2.5. Costs

Costs were expressed in 2020 Euros. Chemotherapy and bevacizumab costs were calculated using the list price [27] and were assessed as the minimum cost per mg of the marketed presentations for each drug. Olaparib and PLD costs were calculated using the confidential price by the SNHS. The first-line maintenance treatment with olaparib was assumed to last up to two years [28]. Since most of the relapsed patients received olaparib as PARPi, the cost of maintenance treatment was assumed to be the same as the olaparib treatment costs in the recurrence setting. Doses and market shares for each drug were also considered (Table S1, S2 and S3). Doses were estimated considering the weight or body surface area of the patients, as indicated in the product labels (Table S3) [28]. Carboplatin dose was determined using the Calvert formula [29]. Du Bois et Du Bois formula was used to calculate the body surface area when necessary [30]. Patients' height and weight were consulted in the Spanish National Health Survey according to the

mean age at diagnosis of stages III and IV [22]. Simulated patients were 54 years old (SD: 10) [32] with a mean height of 161 cm (SD: 5.64) and weight of 67.51 kg (SD: 11.38) [22]. For intravenous drugs, the model also considered the non-optimization of vials and the cost of administration for each drug: cost of administration (€0.57 per min) and time of administration required for each administration in day hospital plus the cost of the 30-min preparation. The administration times were: gemcitabine and bevacizumab (60 min), carboplatin and PLD (90 min) and paclitaxel and trabectedin (210 min) [28].

Unit costs of oncology visits and follow-up tests reflected the median value of the unit costs for all Autonomous Communities in Spain [33]. Adverse events and palliative care costs came from scientific articles regarding the management of cancer (Table 1) [31]. A palliative care cost of € 3701.74 € was applied in the year of the patient's death [26].

Since the costs of the oncology visits, follow-up tests and management of adverse events came from different years, they were updated to 2020 Euros using the medicine consumer price index [22].

2.6. Utilities

Health state utility values were 0.82 in the PFSt, 0.77 in PS1 [13] and 0.68 in PS2 [34]. Health state utilities for advanced HGSOc patients with *BRCA* mutations were also age-adjusted along the time horizon to reflect the aging of the cohort. Mean utilities for the Spanish population were extracted from the Spanish National Health Survey [22] (Table S4).

Disutility associated to adverse events development was also considered in the model (Table 1). It was assumed that the effect of all adverse events on the patients' quality of life lasted 7 days [35].

2.7. Sensitivity analyses

An univariate sensitivity analysis was carried out to examine the simulation's robustness. Different assessments were made based on the possible variation of the most sensitive parameters: the distributions that best fit the Kaplan-Meier survival curves from the clinical trial, according to the Akaike information criterion (AIC) (generalized

Table 1
Frequency, unit costs and disutilities associated to adverse events.

Adverse events	Frequency ¹		Unit cost ²	Disutility
	Olaparib scenario	No maintenance treatment scenario		
Anemia	21.54%	1.54%	€497.43	−0.119 (SD = 0.01) ³
Fatigue or asthenia	3.85%	1.54%	€174.68	−0.073 (SD = 0.018) ⁴
Neutropenia	8.46%	4.62%	€272.10	−0.090 (SD = 0.015) ⁴
Diarrhea	3.08%	0.00%	€1546.37	−0.047 (SD = 0.016) ⁴

Abbreviation: SD: standard deviation. References.

¹ Moore, 2018 [13].

² Isla, 2017 [31].

³ Swinburn, 2010 [44].

⁴ Nafees, 2008 [43].

Table 2
Frequency and unitary costs of the oncology visits and follow-up tests.

Test	Stable patients						Patients showing progression	Unit costs
	Year 1	Year 2	Years 3	Years 4	Year 5	Year 6+		
Physical examination (Oncology visit)	Every 3 months	Every 3 months	Every 6 months	Every 6 months	Every 6 months	Annually	Every 3 months	€88.72
Biochemical analysis and hematology test	Every 3 months	Every 3 months	Every 6 months	Every 6 months	Every 6 months	Annually	Every 3 months	€60.71
Tumor markers (CA-125)	Every 3 months	Every 3 months	Every 6 months	Every 6 months	Every 6 months	Annually	Every 3 months	€11.85
Chest computed tomography	Every 3 months	Every 3 months	Every 6 months	Every 6 months	Every 6 months	Annually	Every 3 months	€207.76 €

References: Autonomous Communities [33], Redondo, 2020 [15].

gamma distribution for first PFS and logistic regression for second PFS and OS), discount rate (0% and 6%), utility in the second-progression state (0.82 QALYs), remission period (7 and 13 years) and the excess mortality risk (0 and 3.42). A probabilistic sensitivity analysis (PSA) was also conducted, including one thousand simulations of the cohort. Most parameters were independently varied according to different distributions and point estimates were drawn using a Monte Carlo simulation (Table S5).

3. Results

3.1. Progression-free and overall survival

The results of the simulation showed that median time to first progression was 4.00 years in patients receiving first-line maintenance treatment with olaparib and 1.50 years in the no maintenance treatment scenario. The average patient receiving first-line maintenance treatment with olaparib had a longer mean average interval until first-progression (5.63 years) than those in the no maintenance treatment scenario (2.06 years). Olaparib patients improved their first-progression survival by 3.57 years in comparison to patients that did not receive maintenance treatment (Table 3).

The median OS in the simulation was 6.50 years and 4.67 years in the olaparib and no maintenance treatment groups, respectively. The model showed that, on average, olaparib patients had a 2.43-year increase in the OS, compared to the no maintenance treatment group (7.26 vs. 4.82, respectively). Therefore, our results suggest that olaparib decreased the mortality risk in more than 30% in advanced HGSC patients with *BRCA* mutations (Table 3).

3.2. Quality adjusted life years

Treatment with Olaparib improved patients' QALYs in the time interval until the first cancer relapse by 2.76 QALYs (4.43 vs. 1.67 QALYs).

Table 3
Cost-effectiveness results per HGSC patient with *BRCA* mutations.

	Olaparib scenario	No maintenance treatment scenario	Difference
Survival(years)			
Period until first progression	5.63	2.06	3.57
Period until second progression	0.67	1.65	−0.98
Overall survival	7.26	4.82	2.43
QALYs			
Period until first progression	4.43	1.67	2.76
Period until second progression	0.51	1.17	−0.66
Total QALYs	5.56	3.57	2.00
Costs (€)			
Treatment			
Olaparib (first line)	71,489.08	0.00	71,489.08
Olaparib (second and subsequent lines)	0.00	31,673.88	−31,673.88
Platinum-based chemotherapy	23,745.15	28,639.15	−4894.0
Non-platinum-based chemotherapy	27,199.73	32,805.74	−5606.01
Total treatment costs	122,433.97	93,118.77	29,315.20
Other healthcare costs			
Period until first progression	3709.75	2198.22	1511.52
Period until second progression	956.83	2197.28	−1240.45
Progression of the disease	1367.08	1591.17	−224.09
Adverse events	184.46	22.90	161.56
Terminal phase	2962.90	3241.20	−278.30
Total other healthcare costs	9181.02	9250.77	−69.75
Total costs	129,320.07	99,601.64	29,718.43
ICER	14,653.20 €		

Abbreviations: PARPi: inhibitor of the poly ADP ribose polymerase; QALYs: quality-adjusted life-years.

These results were in line with the survival results previously registered (Table 3).

In the simulated lifetime of patients, the use of olaparib as a first-line maintenance treatment improved QALYs by 2.00 (5.56 vs. 3.57 QALYs) (Table 3).

3.3. Costs

Treatment costs were higher in patients that received olaparib (€122,433.97) than in the no maintenance treatment scenario (€93,118.77). However, olaparib implied a cost saving in the platinum-based (€4894) and non-platinum-based chemotherapy (€5606.01) administered in the recurrence setting (Table 3).

Regarding other healthcare resources, most of the costs in the olaparib scenario were associated to the period up to the PS1 (40.4%), followed by the terminal phase (32.3%) and the progression of the disease (14.9%). However, the distribution of the costs was different in patients without maintenance treatment, as most of the costs were associated to the terminal phase (35%), followed by the time interval up to the first relapse (23.8%), and the period up to the second relapse (23.8%). As shown in Table 3, the first-line maintenance treatment with olaparib reduced costs associated to the time interval until second progression (€1240.45), the progression disease period (€224.09) and the terminal phase (€278.30) vs. the no maintenance treatment scenario. Therefore, the administration of olaparib to newly diagnosed patients with advanced HGSC and *BRCA* mutations results in health care cost savings of €69.75 per patient, in comparison to the no maintenance treatment scenario (Table 3).

3.4. Incremental cost-effectiveness

Our results showed that, as first-line maintenance treatment, olaparib would lead to an increase of 2.43 years free of disease progression and of €29,245.44 and per patient (Table 3). Therefore, the incremental cost-effectiveness ratio (ICER) was €12,020.67 per free-progression year saved.

Olaparib also implied an improvement in 2.00 QALYs per patient compared to the no maintenance treatment scenario, so the ICER was €14,653.2/QALY. Thus, the administration of olaparib as first-line maintenance treatment is cost-effective, taking into account the cost-effective threshold between €25,000 - €60,000/QALY considered in Spain [20,21].

3.5. Sensitivity analyses

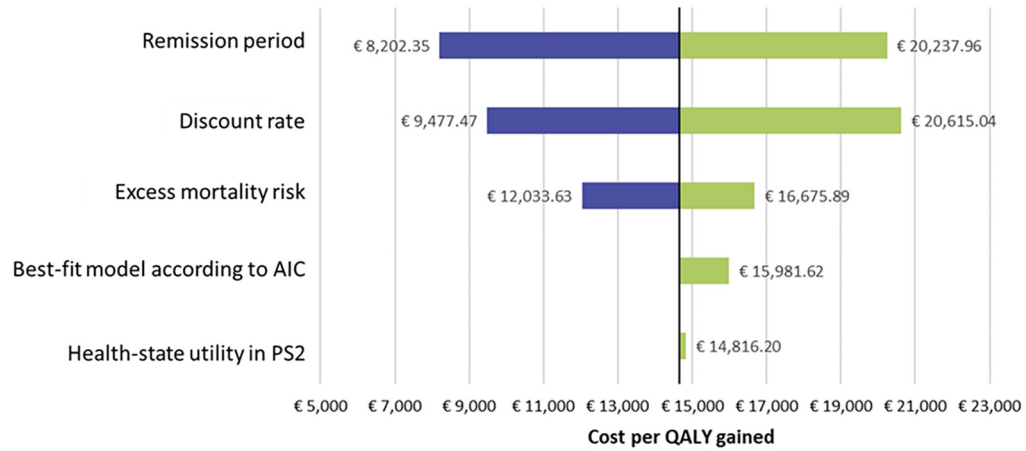
The results of the univariate deterministic analysis showed that the most influential parameters on the cost per QALY gained were the remission period (−44.02%; 38.11%); the discount rate (−35.32%; 40.69%) and the excess mortality risk (−17.88%; 13.80%). The assumed survival distributions (fitting the model according to the AIC) and the variation of the health-state utility in PS2 had a lower impact of cost per QALY gained (9.07% and 1.11%, respectively) (Fig. 2A).

The PSA showed that most of the simulations were located in the upper-right quadrant of the cost-effectiveness plane (Fig. 2B). The ICER ranged from €-9546.19/QALY (dominant) to €51,978.64/QALY. The willingness-to-pay curve is shown in Fig. 2C. The results of the sensitivity analysis showed that 92.1% of the simulations fell below the €25,000/QALY threshold, and 100% were below the €60,000/QALY threshold (Fig. 2C).

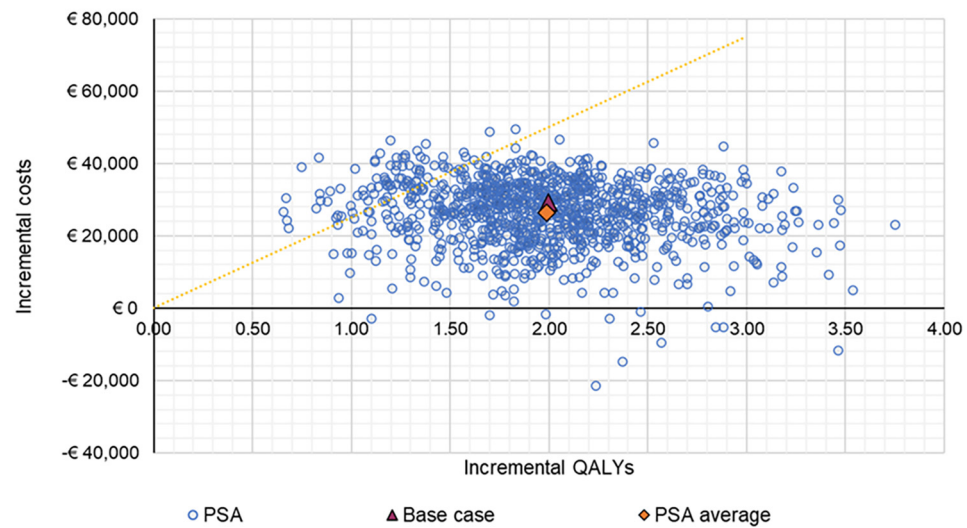
4. Discussion

Treatment with olaparib substantially improving the first PFS (3.57 years), OS (2.43 years) and QALYs (2.00 QALYs) of an average simulated patient with advanced HGSC and *BRCA* mutation for the first-line maintenance treatment. Furthermore, olaparib is cost-effective

A



B



C

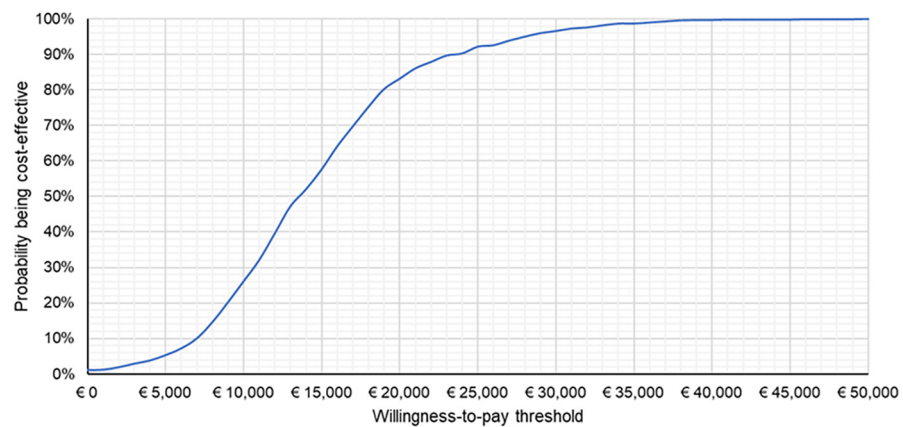


Fig. 2. Sensitivity analyses: tornado diagram (A), cost-effectiveness plane (B) and willingness-to-pay curve (C).

Abbreviations: AIC: Akaike information criterion; PS2: second-progression state; PSA: probabilistic sensitivity analysis; QALYs: quality-adjusted life-years.

(ICER €14,653.2/QALY), taking into account the cost-effective threshold between €25,000 – €60,000/QALY considered in Spain. Our simulation showed that the median first PFS in patients without maintenance treatment was 1.50 years, compared to 4.00 years in the olaparib group, increasing progression free survival by 2.5 years. These results are in line and slightly more conservative than those in the 5-year follow-up from SOLO1 trial, as the median time to first PFS was 13.8 months (1.15 years) and 56 months (4.67 years) in the placebo and olaparib group, respectively [13,17]. In addition, the simulation showed that patients in the no maintenance treatment scenario had a 5-year OS rate of 45%, which is in line with previous clinical evidence [36]. Therefore, our results clearly show that the maintenance treatment with olaparib improves the long-term OS and QALYs of these patients.

To our knowledge, this is one of the first published study that estimates the cost-effectiveness of olaparib as first-line maintenance treatment in Europe. Several economic evaluations were carried out in the United States, regarding the cost-effectiveness of olaparib for the treatment of OC. However, the majority of these evaluations included patients with platinum-sensitive recurrent OC [37–39]. Regarding the recent indication of olaparib in first line, two economic evaluations analyzed the cost-effectiveness of olaparib; one developed in Italy [40] and the other in Singapore [41]. The Italian economic evaluation analyzed the cost-effectiveness of olaparib maintenance therapy compared with no maintenance therapy, according the SOLO1 trial results, from the Italian National Health Service perspective. This study determined that olaparib is cost-effective (€11,345/QALY) as maintenance monotherapy for patients with advanced BRCA1/2-mutated HGSOC who are in response (complete or partial) to first-line platinum-based chemotherapy. A PSA was also developed, which results demonstrated that olaparib was associated with a 97.6% probability of being cost-effective, considering a threshold of €16,372/QALY [40]. Our results are in line with these conclusions, as 92.41% of the simulations from our probabilistic sensitivity analysis fell below the €25,000/QALY threshold, which is considered cost-effective as per the Spanish guidelines [20,21].

The economic evaluation developed in Singapore analyzed the cost-effectiveness of olaparib maintenance therapy versus routine surveillance, according to the results from the SOLO1 trial. This study was developed from the healthcare payer perspective, using similar conditions to the ones we considered in our study. It showed that olaparib maintenance vs. routine surveillance resulted in an ICER of SGD17,326/QALY (around €11,500/QALY). The researchers also developed a PSA, which demonstrated that olaparib was associated with an 81% probability of being cost-effective, considering a threshold of SGD50,000/QALY (around €33,000/QALY). In addition, their results were most sensitive to variations in the discount rate and mortality risk in long-term survivors [41]. These conclusions are in agreement with our results [20,21]. It should be noted that researchers considered a remission period of 7 years, while we assumed 10 years in the base case. A 7-year remission period was included in our sensitivity analysis, resulting in an ICER of €9202.35/QALY, lower than our base case results, and closer to the ICER in Singapore. In addition, it should be stressed that the two last studies simulated patients over a three-state model; however, we considered four states, based on the first-progression and second-progression free survival from the SOLO1 trial. We estimate that our four-state model is more accurate to simulate the outcomes of the olaparib treatment in these patients.

Our study has some limitations. First, due to the lack of information about the real-world outcomes of the OC treatment in Spanish patients with advanced HGSOC and BRCA mutation, the model included the results from the SOLO1 trial (efficacy and utilities) which are selected patients that were required to accomplish several criteria of inclusion. Second, based on data from SOLO1, the simulation did not include either the antiangiogenic therapy (bevacizumab) as first-line maintenance treatment nor the treatment with other PARPi for recurrent HGSOC. It should be noted that the Italian study collected these treatments

obtaining results similar to ours [40]. Third, although the efficacy results available from SOLO1 were still immature, our model considered those for the assessments, including the OS (data maturity, 21%). Four, we took into account the burden associated to all severe adverse events (grades 3 and 4) registered in the SOLO1 trial, in terms of impact on health status and costs. Even if we assume that most of the burden is indeed associated to severe adverse events, those graded 1 and 2 may be also relevant in the determination of quality of life and costs, and our model may be underestimating the total burden. Moreover, we did not consider the development of adverse events due to the second and subsequent lines of treatment. Finally, costs and disutilities associated to adverse events came from studies developed in other cancers. However, the variations in base-case values, including the duration of the adverse events, were analyzed in the PSA. Finally, although the resources use (oncology visits and follow-up tests) was agreed by a multidisciplinary expert group, the real-world practice in some Spanish hospitals may be different (e.g., pelvic ultrasound may also be included in the follow-up procedures). Therefore, these possible variations were evaluated in the PSA, and the corresponding results were compared to the cost-utility threshold.

Despite its limitations, the main contribution of this study is the economic evaluation of olaparib for the maintenance therapy of newly diagnosed patients with advanced HGSOC and BRCA mutation in Spain. Our results confirm that the incremental cost associated with olaparib monotherapy administered to patients in response after first-line platinum-based chemotherapy is associated with increased overall survival and QALYs, and is a cost-effective treatment option in Spain. These results are in line with recent studies, concluding that maintenance treatment with targeted therapies in advanced OC patients can be cost-effective [42]. In conclusion, our results support the strong recommendation of SEOM guidelines [15] about the use of olaparib as first-line maintenance treatment in patients with advanced HGSOC BRCA mutated- with a cost-effective ICER of €14,653.2/QALY, according to the cost-effectiveness threshold considered in Spain [20,21].

Conflict of interest statement

VGB reports personal fees from Astra Zeneca, during the conduct of the study and personal fees from GSK, outside the present study. MPBG reports fees from GlaxoSmithKline, Clovis Oncology, PharmaMar, AstraZeneca, MSD, and Roche for consulting or advisory functions and travel support from GlaxoSmithKline, PharmaMar, Astra Zeneca, MSD, and Roche, outside the present study. JAPF reports personal fees from AstraZeneca, Clovis, Roche, Abilify Pharma and Pharmamar for speaker bureau or advisory functions and grants and personal fees from GSK for speaker bureau, advisory functions and research grant, outside the present study. AR reports grants and personal fees from Pharmamar and Roche for research funding, travel expenses and advisory functions, personal fees from Amgen, Astra Zeneca, GSK and Clovis for advisory functions, travel expenses or speakers bureau, and grants from Eisai for research funding, outside the present study. CMA is an AstraZeneca employee. AGD and YIM work at Weber, company that received fees from AstraZeneca, during the conduct of the study.

Author contribution

CMA and AGD conceived and designed this study. All authors contributed to the acquisition of the data. AGD contribute to analysis of the data. All authors contributed to interpretation of the data. AGD and YIM contributed to the development of the drafts of this manuscript. CMA, VGB, MPBG, JAPF and AR critically revised the manuscript. All authors approved the submitted version of the manuscript.

Acknowledgement

The authors would like to thank Rob Hettle for developing the model for the United Kingdom that served as the basis for conducting this research.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ygyno.2021.11.011>.

References

- [1] Sociedad Española de Oncología Médica (SEOM), Las cifras del cáncer en España, 2021 https://seom.org/images/Cifras_del_cancer_en_Espana_2021.pdf (accessed April 27, 2021).
- [2] M. Matz, M.P. Coleman, H. Carreira, D. Salmerón, M.D. Chirilaque, C. Allemani, World-wide comparison of ovarian cancer survival: histological group and stage at diagnosis (CONCORD-2), *Gynecol. Oncol.* 144 (2017) 396–404, <https://doi.org/10.1016/j.ygyno.2016.11.019>.
- [3] I. Romero, R.C. Bast, Minireview: human ovarian Cancer: biology, current management, and paths to personalizing therapy, *Endocrinology*. 153 (2012) 1593–1602, <https://doi.org/10.1210/en.2011-2123>.
- [4] S. González-Santiago, T. Ramón Y. Cajal, E. Aguirre, J.E. Alés-Martínez, R. Andrés, J. Balmaña, B. Graña, A. Herrero, G. Llort, A. González-del-Alba, The SEOM hereditary Cancer working group, SEOM clinical guidelines in hereditary breast and ovarian cancer (2019), *Clin. Transl. Oncol.* 22 (2020) 193–200, <https://doi.org/10.1007/s12094-019-02262-0>.
- [5] The Cancer Genome Atlas Research Network, Integrated genomic analyses of ovarian carcinoma, *Nature*. 474 (2011) 609–615, <https://doi.org/10.1038/nature10166>.
- [6] E. Capoluongo, G. Scambia, J.-M. Nabholz, Main implications related to the switch to BRCA1/2 tumor testing in ovarian cancer patients: a proposal of a consensus, *Oncotarget*. 9 (2018) 19463–19468, <https://doi.org/10.18632/oncotarget.24728>.
- [7] A. Oaknin, R. Guarch, P. Barretina, D. Hardisson, A. González, X. Matías-Guiu, A. Pérez-Fidalgo, B. Veiates, I. Romero, J. Palacios, Recommendations for biomarker testing in epithelial ovarian cancer: a National Consensus Statement by the Spanish Society of Pathology and the Spanish Society of Medical Oncology, *Clin. Transl. Oncol.* (2017), <https://doi.org/10.1007/s12094-017-1719-x>.
- [8] S. Narod, Can advanced-stage ovarian cancer be cured? *Nat. Rev. Clin. Oncol.* 13 (2016) 255–261, <https://doi.org/10.1038/nrclinonc.2015.224>.
- [9] J.A. Ledermann, First-line treatment of ovarian cancer: questions and controversies to address, *Ther. Adv. Med. Oncol.* 10 (2018), <https://doi.org/10.1177/1758835918768232>.
- [10] B.M. Norquist, M.F. Brady, M.I. Harrell, T. Walsh, M.K. Lee, S. Gulsner, S.S. Bernards, S. Casadei, R.A. Burger, K.S. Tewari, F. Backes, R.S. Mannel, G. Glaser, C. Bailey, S. Rubin, J. Soper, H.A. Lankes, N.C. Ramirez, M.C. King, M.J. Birrer, E.M. Swisher, Mutations in homologous recombination genes and outcomes in ovarian carcinoma patients in GOG 218: an NRG oncology/gynecologic oncology group study, *Clin. Cancer Res.* 24 (2018) 777–783, <https://doi.org/10.1158/1078-0432.CCR-17-1327>.
- [11] L.C. Hanks, S. Loibl, N. Burchardi, J. Pfisterer, W. Meier, E. Pujade-Lauraine, I. Ray-Coquard, J. Sehouli, P. Harter, A. du Bois, AGO and GINECO study group, the impact of second to sixth line therapy on survival of relapsed ovarian cancer after primary taxane/platinum-based therapy, *Ann. Oncol.* 23 (2012) 2605–2612, <https://doi.org/10.1093/annonc/mds203>.
- [12] C. Fotopoulou, Limitations to the use of carboplatin-based therapy in advanced ovarian cancer, *EJC Suppl.* 12 (2014) 13–16, [https://doi.org/10.1016/S1359-6349\(15\)70005-4](https://doi.org/10.1016/S1359-6349(15)70005-4).
- [13] K. Moore, N. Colombo, G. Scambia, B.-G. Kim, A. Oaknin, M. Friedlander, A. Lisyanskaya, A. Floquet, A. Leary, G.S. Sonke, C. Gourley, S. Banerjee, A. Oza, A. González-Martín, C. Aghajanian, W. Bradley, C. Mathews, J. Liu, E.S. Lowe, R. Bloomfield, P. DiSilvestro, Maintenance Olaparib in patients with newly diagnosed advanced ovarian cancer, *N. Engl. J. Med.* 379 (2018) 2495–2505, <https://doi.org/10.1056/NEJMoa1810858>.
- [14] National Comprehensive Cancer Network (NCCN), NCCN Clinical practice guidelines in oncology. Ovarian cancer including Fallopian tube cancer and primary peritoneal cancer, 2019.
- [15] A. Redondo, E. Guerra, L. Manso, C. Martín-Lorente, J. Martínez-García, J.A. Pérez-Fidalgo, M.Q. Varela, M.J. Rubio, M.P. Barretina-Ginesta, A. González-Martín, SEOM clinical guideline in ovarian cancer (2020), *Clin. Transl. Oncol.* (2021), <https://doi.org/10.1007/s12094-020-02545-x>.
- [16] L. Delgado-Ortega, J. Ginés Rubió, M.D.C. Garcías de España, D. Carcedo, L. Cordero Puentes, C. Moya de Alarcón, Economic impact of olaparib on maintenance treatment of patients with BRCA-mutation positive, platinum-sensitive relapsing high-grade serous epithelial ovarian cancer in Spain, *Farm. Hosp.* 42 (2018) 95–102, <https://doi.org/10.7399/fh.10904>.
- [17] S. Banerjee, K.N. Moore, N. Colombo, G. Scambia, B.-G. Kim, A. Oaknin, M. Friedlander, A. Lisyanskaya, A. Floquet, A. Leary, G.S. Sonke, C. Gourley, A. Oza, A. González-Martín, C. Aghajanian, W.H. Bradley, E. Holmes, E.S. Lowe, P. DiSilvestro, Maintenance olaparib for patients with newly diagnosed advanced ovarian cancer and a BRCA mutation (SOLO1/GOG 3004): 5-year follow-up of a randomised, double-blind, placebo-controlled, phase 3 trial, *Lancet Oncol.* (2021), [https://doi.org/10.1016/S1473-2045\(21\)00531-3](https://doi.org/10.1016/S1473-2045(21)00531-3).
- [18] National Institute for Health and Care Excellence (NICE), Olaparib for maintenance treatment of BRCA mutation-positive advanced ovarian, fallopian tube or peritoneal cancer after response to first-line platinum-based chemotherapy. Technology appraisal guidance [TA598], 2019, <https://www.nice.org.uk/guidance/ta598>.
- [19] J.L. Bastida, J. Oliva, F. Antónanzas, A. García-Altés, R. Gisbert, J. Mar, J. Puig-Junoy, Propuesta de guía para la evaluación económica aplicada a las tecnologías sanitarias, *Gac. Sanit.* 24 (2010) 154–170.
- [20] J.A. Sacristán, J. Oliva, C. Campillo-Artero, J. Puig-Junoy, J.L. Pinto-Prades, T. Dilla, C. Rubio-Terrés, V. Ortún, ¿Qué es una intervención sanitaria eficiente en España en 2020? [What is an efficient health intervention in Spain in 2020?], *Gac. Sanit.* 34 (2019) 189–193, Spanish <https://doi.org/10.1016/j.gaceta.2019.06.007>.
- [21] L. Vallejo-Torres, B. García-Lorenzo, P. Serrano-Aguilar, Estimating a cost-effectiveness threshold for the Spanish NHS, *Health Econ.* 27 (2018) 746–761.
- [22] Instituto Nacional de Estadística. (Spanish Statistical Office), (N.d.), <http://www.ine.es/en/welcome.shtml> 2021.
- [23] P.L. Mai, N. Chatterjee, P. Hartge, M. Tucker, L. Brody, J.P. Struwing, S. Wacholder, Potential excess mortality in BRCA1/2 mutation carriers beyond breast, ovarian, prostate, and pancreatic cancers, and melanoma, *PLoS One* 4 (2009), <https://doi.org/10.1371/journal.pone.0004812>.
- [24] K.G. Essel, K. Behbakht, T. Lai, E. Evans, J.D. Dvorak, K. Ding, G.E. Konecny, K.N. Moore, PARPi after PARPi in epithelial ovarian cancer, *Gynecol. Oncol.* 149 (2018) 176–177, <https://doi.org/10.1016/j.ygyno.2018.04.401>.
- [25] European Medicines Agency, European public assessment report (EPAR) for Lynparza, https://www.ema.europa.eu/en/documents/product-information/lynparza-epar-product-information_en.pdf 2019 (accessed September 20, 2019).
- [26] C. Moya-Alarcón, A. González-Domínguez, S. Simon, I. Pérez-Román, A. González-Martín, E. Bayo-Lozano, A.B. Sánchez-Heras, Cost-utility analysis of germline BRCA1/2 testing in women with high-grade epithelial ovarian cancer in Spain, *Clin. Transl. Oncol.* 21 (2019) 1076–1084, <https://doi.org/10.1007/s12094-018-02026-2>.
- [27] Consejo General de Colegios Oficiales de Farmacéuticos, Bot Plus 2.0 Base de Datos de Medicamentos, <https://botplusweb.portalafarma.com/botplus.aspx> 2021 (accessed January 21, 2021).
- [28] Agencia Española de Medicamentos y Productos Sanitarios (AEMPS), Centro de información de medicamentos (CIMA), 2020 <https://cima.aemps.es/cima/publico/home.html> (accessed February 9, 2021).
- [29] A.H. Calvert, D.R. Newell, L.A. Gumbrell, S. O'Reilly, M. Burnell, F.E. Boxall, Z.H. Siddik, I.R. Judson, M.E. Gore, E. Wiltshaw, Carboplatin dosage: prospective evaluation of a simple formula based on renal function, *J. Clin. Oncol.* 7 (1989) 1748–1756.
- [30] D. Du Bois, E.F. Du Bois, A formula to estimate the approximate surface area if height and weight be known. 1916, *Nutrition* 5 (1989) 303–311 (discussion 312–313).
- [31] D. Isla, J. De Castro, O. Juan, S. Grau, J. Orfino, R. Gordo, C. Rubio-Terrés, D. Rubio-Rodríguez, Costs of adverse events associated with erlotinib or afatinib in first-line treatment of advanced EGFR-positive non-small cell lung cancer, *Clinicoecon Outcomes Res.* 9 (2017) 31–38, <https://doi.org/10.2147/CEOR.S121093>.
- [32] F.J. Candido-dos-Reis, H. Song, E.L. Goode, J.M. Cunningham, B.L. Fridley, M.C. Larson, K. Alsop, E. Dicks, P. Harrington, S.J. Ramus, A. de Fazio, G. Mitchell, S. Fereday, K.L. Bolton, C. Gourley, C. Michie, B. Karlan, J. Lester, C. Walsh, I. Cass, H. Olsson, M. Gore, J.J. Benitez, M.J. Garcia, I. Andrusis, A.M. Mulligan, G. Glendon, I. Blanco, C. Lazarro, A.S. Whittemore, V. McGuire, W. Sieh, M. Montagna, E. Alducci, S. Sadetzki, A. Chetrit, A. Kwong, S.K. Kjaer, A. Jensen, E. Høgdall, S. Neuhausen, R. Nussbaum, M. Daly, M.H. Greene, P.L. Mai, J.T. Loud, K. Moysich, A.E. Toland, D. Lambrechts, S. Ellis, D. Frost, J.D. Brenton, M. Tischkowitz, D.F. Easton, A. Antoniou, G. Chenevix-Trench, S.A. Gayther, D. Bowtell, P.D.P. Pharoah, Germline mutation in BRCA1 or BRCA2 and ten-year survival for women diagnosed with epithelial ovarian cancer, *Clin. Cancer Res.* 21 (2015) 652–657, <https://doi.org/10.1158/1078-0432.CCR-14-2497>.
- [33] Median value of the unit costs for autonomous, Communities. (2020).
- [34] National Institute for Health and Care Excellence (NICE), Single Technology Appraisal. Olaparib for maintenance treatment of relapsed platinum-sensitive ovarian, fallopian tube or peritoneal cancer, Technology appraisal guidance [TA620] (2020), Committee Papers <https://www.nice.org.uk/guidance/indevelopment/gid-ta10303/documents> (accessed December 17, 2019).
- [35] Expert group, (2019).
- [36] Y. D. K. S. Y. H. K. S. I. S. Ak, Z. W. Association of BRCA1 and BRCA2 mutations with survival, chemotherapy sensitivity, and gene mutator phenotype in patients with ovarian cancer, *JAMA*. 306 (2011) 1557–1565, <https://doi.org/10.1001/jama.2011.1456>.
- [37] L. Zhong, Cost-Effectiveness of Niraparib and Olaparib as Maintenance Therapy for Patients with Platinum-Sensitive Recurrent Ovarian Cancer, 10, 2021.
- [38] H. Guy, L. Walder, M. Fisher, Cost-effectiveness of Niraparib versus routine surveillance, Olaparib and Rucaparib for the maintenance treatment of patients with ovarian Cancer in the United States, *Pharmacoeconomics*. 37 (2019) 391–405, <https://doi.org/10.1007/s40273-018-0745-z>.
- [39] H.J. Smith, C.L. Walters Haygood, R.C. Arend, C.A. Leath, J.M. Straughn, PARP inhibitor maintenance therapy for patients with platinum-sensitive recurrent ovarian cancer: a cost-effectiveness analysis, *Gynecol. Oncol.* 139 (2015) 59–62, <https://doi.org/10.1016/j.ygyno.2015.08.013>.
- [40] P. Armeni, L. Borsoi, G. Fornaro, C. Jommi, N. Colombo, F. Costa, Cost-effectiveness and net monetary benefit of Olaparib maintenance therapy versus no maintenance therapy after first-line platinum-based chemotherapy in newly diagnosed advanced BRCA1/2-mutated ovarian Cancer in the Italian National Health Service, *Clin. Ther.* 42 (2020) 1192–1209.e12, <https://doi.org/10.1016/j.clinthera.2020.04.015>.
- [41] D.S. Tan, J.J. Chan, J.K. Loke, R. Hettle, W. Ghosh, A. Viswambaram, C.C. Yu, Cost-effectiveness of olaparib vs routine surveillance in the maintenance setting for patients with BRCA-mutated advanced ovarian cancer after response to first-line

- platinum-based chemotherapy in Singapore, *Ann. Oncol.* 30 (2019), <https://doi.org/10.1093/annonc/mdz426.007> ix79–ix80.
- [42] M.T. Richardson, S.-Y. Liang, D.S. Kapp, J.E. Chan, A.K. Mann, B.J. Monk, T.J. Herzog, D.S. Lakomy, R.L. Coleman, J.K. Chan, Upfront maintenance with anti-vascular versus PARP inhibitors in advanced ovarian cancer: a cost-effective analysis, 2020, <https://sgo.confex.com/sgo/2020/meetingapp.cgi/Paper/15316>.
- [43] B. Nafees, M. Stafford, S. Gavriel, S. Bhalla, J. Watkins, Health state utilities for non small cell lung cancer, *Health Qual. Life Outcomes* 6 (2008) 84, <https://doi.org/10.1186/1477-7525-6-84>.
- [44] P. Swinburn, A. Lloyd, P. Nathan, T.K. Choueiri, D. Cella, M.P. Neary, Elicitation of health state utilities in metastatic renal cell carcinoma, *Curr. Med. Res. and Opin. Curr. Med. Res. and Opin.* 26 (2010) 1091–1096, <https://doi.org/10.1185/03007991003712258>.